Polarized Discourse on AI-Generated Art: Constructing a Graph Benchmark from Reddit for Stance Classification

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1 Introduction

Online social networks have become one of the main spaces where public debates unfold, especially around recent and controversial topics. Understanding how individuals argue and interact around polarized topics on these platforms has important implications for the study of misinformation, content moderation, and political and cultural polarization. While political polarization on platforms such as Twitter and Reddit has been extensively studied, cultural and technological polarization, particularly around emerging AI capabilities, remains underexplored from a graph-based computational perspective.

The rapid proliferation of generative AI tools such as Stable Diffusion, Midjourney, and DALL-E has prompted us to focus on the debate: *Should AI-generated art be banned?*. Driven by concerns about intellectual property, plagiarism, artistic labor, and the use of training data, this topic has generated substantial discourse on social media and makes a compelling case study for understanding online polarization.

These debates are structured through interactions between users, not just isolated opinions. On Reddit, users cluster around shared viewpoints and engage through replies, disagreement, and selective attention (Quattrociocchi et al., 2016). The objective of this project is therefore to construct a small, interpretable, and reproducible graph benchmark dataset and evaluate whether incorporating interaction structure improves stance classification performance compared to text-only approaches.

2 Related Work

2.1 Polarization on social networks

Political polarization online has been studied extensively. Conover et al. (2011) showed that retweet networks on Twitter split cleanly along partisan lines, while Hofmann et al. (2022) showed that graph-based models capture framing patterns in polarized Reddit groups that text alone may miss. Stance detection has also been framed primarily as a text classification task, with methods ranging from logistic regression to fine-tuned transformers (Augenstein et al., 2016). However, this work has largely focused on political divides and usually treats comments as independent rather than socially connected.

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2.2 AI art sentiment and graph-based methods

Recent studies explore public attitudes toward AI-generated art, identifying recurring concerns such as plagiarism, copyright, artistic labor, and job displacement (Wang et al., 2025; Ali et al., 2023; Miyazaki et al., 2024). GNNs have been applied to similar problems in social media, including thread-level classification (Zheng et al., 2025). However, graph-based methods have not yet been applied to stance classification in the AI art debate, where reply structure may be informative even when interactions are not homophilous.

Overall, polarization has been studied on graphs, but only for political topics. AI art sentiment has been studied, but only with text methods. Graph benchmarks exist, but none target culturally polarized discourse. We sit at the intersection of these three gaps.

3 Problem Definition

We formulate the problem as a semi-supervised node classification task on a user-interaction graph. We model the discussion as a directed graph $G = (V, E)$, where nodes represent Reddit users and a directed edge (u, v) exists if user u replied to a post or comment by user v.

Each node is associated with a feature vector derived from the user's comments. We encode each comment with a pretrained Sentence-BERT model (`all-MiniLM-L6-v2`) and average the resulting embeddings per user to produce a single feature vector. Each labelled user node gets a binary stance label: 1 for supporting a ban on AI-generated art, and 0 for opposing one. Ambiguous, neutral, or off-topic users are left unlabelled but kept in the graph so that their interaction patterns can still contribute structural context for graph-based models.

Given the full graph structure, the task is to learn a classifier $f : V \to \{0, 1\}$ that predicts the stance of unlabelled users. We compare a text-only MLP against three GNN models: GCN (Kipf and Welling, 2017), GAT (Veličković et al., 2018), and GraphSAGE (Hamilton et al., 2017). We evaluate performance on the labelled subset using accuracy, macro F1-score, and per-class precision / recall on a stratified 60/20/20 train/validation/test split.

We assume that reply interactions encode social signals that are informative for stance classification, whether through agreement, opposition, or local interaction structure.

4 Dataset

The dataset is constructed from Reddit threads discussing whether AI-generated art should be banned. We will select $3 - 6$ threads that explicitly centre on this debate, prioritising broad-discussion and debate-oriented subreddits while keeping the total dataset small enough to remain feasible for manual annotation.

From each thread, we extract user identifiers, comment text, and reply relations between comments. Users are represented as nodes and replies as directed edges, and the collected threads are combined into a single graph. We expect the dataset to contain around $500 - 1,000$ comments, providing enough interaction structure to support graph-based modelling without making annotation unmanageable.

The final dataset will be represented by an edge list for the graph structure, a node feature matrix derived from user comments, and a partially observed binary label vector over the labelled subset. Reddit suits this task because threaded discussions provide reply structure directly, allowing us to model not only what users say but also how they interact.

Limitations include that the dataset is relatively small, stance annotation is inherently somewhat subjective, and Reddit users are not representative of the general population.

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